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**Narula Institute of Technology**  
**An Autonomous Institute under MAKAUT**  
**2023**  
**END SEMESTER EXAMINATION - ODD 2023**  
**HU104 - ENVIRONMENTAL SCIENCE**

**TIME ALLOTTED: 3Hours****FULL MARKS: 70**

*Instructions to the candidate:*

*Figures to the right indicate full marks.*

*Draw neat sketches and diagram wherever is necessary.*

*Candidates are required to give their answers in their own words as far as practicable*

**Group A**  
**(Multiple Choice Type Questions)**

**Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. Food Web is (1) CO1 BL2
  - a) Single food chain
  - b) Disintegrated food chain
  - c) Interconnected food chain
  - d) None
2. Example of renewable resource (1) CO1 BL1
  - a) both Solar Energy and Wind Energy
  - b) Solar Energy
  - c) Wind Energy
  - d) Fossil fuel
3. Minamata disease is caused due to presence of (1) CO4 BL2
  - a) Pb
  - b) As
  - c) Hg
  - d) Cd
4. Hazardous waste should have characteristics as (1) CO4 BL3
  - a) Toxicity
  - b) Corrosivity
  - c) Reactivity
  - d) All of these
5. Ozone is pollutant when present in (1) CO1 BL2
  - a) Mesosphere
  - b) Stratosphere
  - c) Troposphere
  - d) Ionosphere

6. Which among of the following is not an example of natural calamity (1) CO2 BL2  
 a) Earthquake  
 b) Flood  
 c) Landslide  
 d) Oil Spill
7. Water will be considered as Saline water if the TDS value is (1) CO4 BL1  
 a) = 1500mg/lit  
 b) less than 500mg/lit  
 c) >2100mg/lit  
 d) None
8. A commonly used coagulant used for waste water treatment is (1) CO2 BL1  
 a) Alum  
 b) Trickling Filter  
 c) Molecular sieves  
 d) Calcium Bicarbonate
9. Montreal Protocol is directly related to? (1) CO3 BL1  
 a) Ozone Layer Depletion  
 b) Green House Effect  
 c) Sustainable development  
 d) E-waste management
10. Temporary hardness of water is due to (1) CO1 BL1  
 a)  $\text{NO}_3^-$   
 b)  $\text{Cl}^-$   
 c)  $\text{HCO}_3^-$   
 d)  $\text{SO}_4^{2-}$
11. Greenhouse effect is due to (1) CO2 BL1  
 a) Over cultivation of land  
 b) Testing nuclear weapons  
 c) Some atmospheric gases like  $\text{CO}_2$ ,  $\text{H}_2\text{O}$  vapour and some manmade gases  
 d) None of these
12. The disinfection by chlorine is due to the formation of (1) CO2 BL1  
 a) Chlorine radical  
 b) Nascent oxygen  
 c) Oxygen gas  
 d) None of these

**Group B**  
**(Short Answer Type Questions)**  
**(Answer any three of the following) 3x5=15**

- 13a. Mention the consequences of global warming (3) CO4 BL2
- 13b. What is Doubling Time of population ? (2) CO1 BL3
14. Write down two examples for each type of solid waste: i) Domestic, (5) CO4 BL3  
 ii) Biomedical, iii) Industrial, iv) Agricultural, and v) E-Waste

- 15a. A waste water sample of 50 ml is diluted by adding 100 ml dilution water. DO<sub>i</sub> is found to be 6 mg/l and after five days the final DO<sub>f</sub> is 3 mg/l. Find the BOD<sub>5</sub> of the waste. (3) CO4 BL4
- 15b. What do you mean by permanent and temporary hardness of water? (2) CO4 BL2
16. Illustrate the difference between Primary and Secondary Air Pollutants with suitable examples. (5) CO2 BL2
17. Write down the differences between renewable and non renewable energy resources with suitable example. (5) CO1 BL2

### Group C

#### (Long Answer Type Questions)

(Answer any three of the following) 3x15=45

- 18a. Write down cause, effects and control of Acid Rain. (5) CO4 BL3
- 18b. Distinguish between Photochemical smog and Sulphurous smog (5) CO4 BL2
- 18c. How do we control air pollution by using cyclone separator? (5) CO4 BL3
19. Answer all the questions (15)
- a) Discuss the process of ozone layer depletion and the photochemical reactions involving in it. What is the impact of ozone layer depletion in human civilization? (5) CO2 BL2
- b) World population in 1850 has been estimated to have been about 1 million. It reached 4 million in 1975. Use the exponential growth rate equation to find the growth rate constant (R) and calculate the doubling time ( $t_d$ ) value. (5) CO4 BL3
- c) Explain bio-magnifications. What is its significance in food chain? (5) CO1 BL2
20. Write short notes on the following (15)
- a) Kyoto protocol (5) CO3 BL2
- b) Determination of Cl<sup>-</sup> ion concentration in water sample (5) CO2 BL2
- c) Municipal Solid waste management (5) CO2 BL2
21. Answer all the questions (15)
- a) What is Global warming? What are the impacts of Global warming on Global climate change? (5) CO3 BL2
- b) What will the ratio of BOD<sub>5</sub> at 20 °C, to that of BOD<sub>2.5</sub> at 35 °C? (5) CO4 BL3
- c) Write down the differences between Sulphurous SMOG and Photochemical SMOG. (5) CO2 BL2
22. Answer all the questions (15)
- a) Write down the names of the heavy metal ions/ anions that are responsible for: Minamata Disease, Black Foot Disease, Blue Baby Syndrome, Itai-Itai, Anemia (5) CO1 BL1

b) Calculate the temporary and permanent hardness of a water sample (5) CO4 BL3  
from the following data:

$\text{Ca}(\text{HCO}_3)_2 = 15 \text{ mg/L}$

$\text{MgCl}_2 = 10.5 \text{ mg/L}$

c) What is Incineration? How it is related to bio-medical waste (5) CO2 BL2  
management?

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